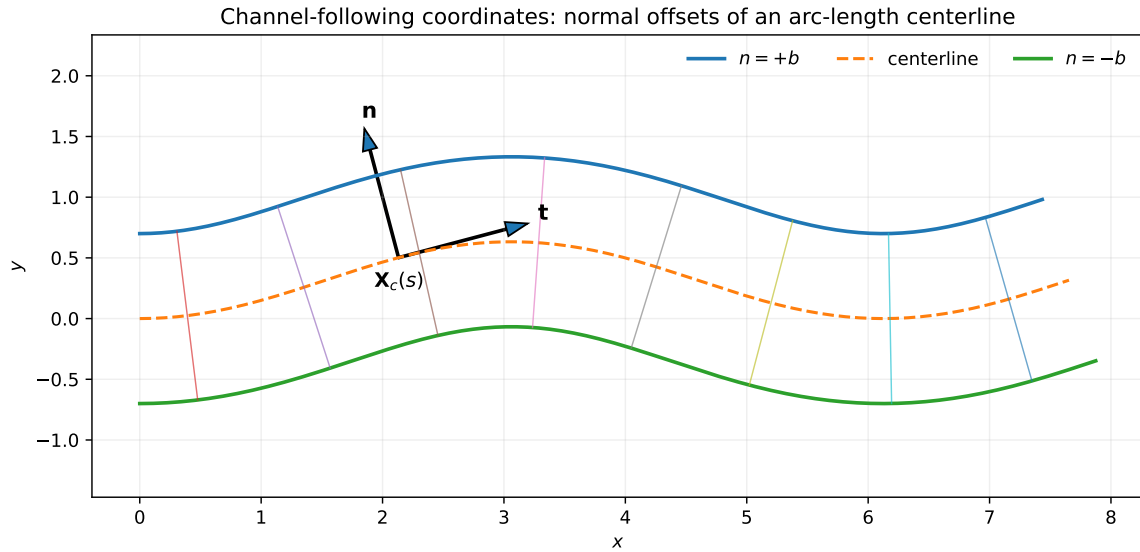


Derivation of a Channel-Following QGPV Equation

for a Meandering River Channel

Exact metric terms, linearization about a prescribed shear profile, and the weak-curvature approximation



Prepared July 2026

Sign convention: $u_s = -\psi_n$, $u_n = h^{-1}\psi_s$

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1 Scope and model hierarchy

This document derives a two-dimensional channel-following barotropic potential-vorticity equation for a fixed meandering channel. The principal result is the exact linearized equation for perturbations about a prescribed along-channel velocity profile $U(n)$, followed by a systematic first-order weak-curvature approximation.

The derivation is organized in the following hierarchy:

1. Define an orthogonal coordinate system (s, n) based on centerline arc length s and signed normal distance n .
2. Derive the metric, gradient, divergence, Laplacian, vorticity, and Jacobian exactly.
3. Write the exact nonlinear barotropic PV equation.
4. Introduce a prescribed base flow $U(n)$ and compute its exact geometry-dependent vorticity gradients.
5. Linearize the PV equation while retaining every metric term.
6. Expand the exact equation to first order in the small parameter $\epsilon_c = b \max |\kappa|$.
7. Specialize to sinusoidal curvature $\kappa(s) = \kappa_0 \cos(Ks)$ and the parabolic profile used in the motivating equation.

The core derivation uses the barotropic limit

$$q = \zeta = \nabla_g^2 \psi, \quad (1)$$

which is the curved-channel analogue of $q = \nabla^2 \psi$ in a straight Cartesian channel. A finite deformation radius, planetary vorticity, and topographic PV can be added afterward; Section 13 gives the corresponding modification.

Notation warning. The symbol κ denotes signed centerline curvature, while K denotes the wavenumber of a prescribed curvature pattern. They have different meanings and dimensions.

2 Channel geometry and Frenet coordinates

2.1 Centerline, tangent, normal, and curvature

Let the channel centerline be a plane curve

$$\mathbf{X}_c(s) = (x_c(s), y_c(s)), \quad (2)$$

parameterized by arc length s . Arc-length parameterization implies

$$\left| \frac{d\mathbf{X}_c}{ds} \right| = 1. \quad (3)$$

Define the unit tangent and the left-pointing unit normal by

$$\mathbf{t}(s) = \frac{d\mathbf{X}_c}{ds} = (\cos \theta(s), \sin \theta(s)), \quad (4)$$

$$\mathbf{n}(s) = (-\sin \theta(s), \cos \theta(s)). \quad (5)$$

The signed curvature is

$$\kappa(s) = \frac{d\theta}{ds}. \quad (6)$$

With this orientation, the Frenet relations are

$$\frac{d\mathbf{t}}{ds} = \kappa\mathbf{n}, \quad \frac{d\mathbf{n}}{ds} = -\kappa\mathbf{t}. \quad (7)$$

Positive κ means that the centerline turns toward the chosen left normal.

2.2 Normal-offset coordinate map

Introduce the channel-following map

$$\mathbf{X}(s, n) = \mathbf{X}_c(s) + n\mathbf{n}(s), \quad (8)$$

where n is signed physical distance normal to the centerline. For a constant half-width b , the banks are the coordinate lines

$$n = +b, \quad n = -b. \quad (9)$$

Differentiating Equation (8) and using Equation (7) gives

$$\mathbf{X}_s = \mathbf{t} + n\mathbf{n}_s = \mathbf{t} - n\kappa\mathbf{t} = (1 - \kappa n)\mathbf{t}, \quad (10)$$

$$\mathbf{X}_n = \mathbf{n}. \quad (11)$$

Therefore the orthogonal scale factors are

$$h_s = h(s, n) = 1 - \kappa(s)n, \quad h_n = 1. \quad (12)$$

The metric and area element are

$$d\ell^2 = h^2 ds^2 + dn^2, \quad dA = h ds dn. \quad (13)$$

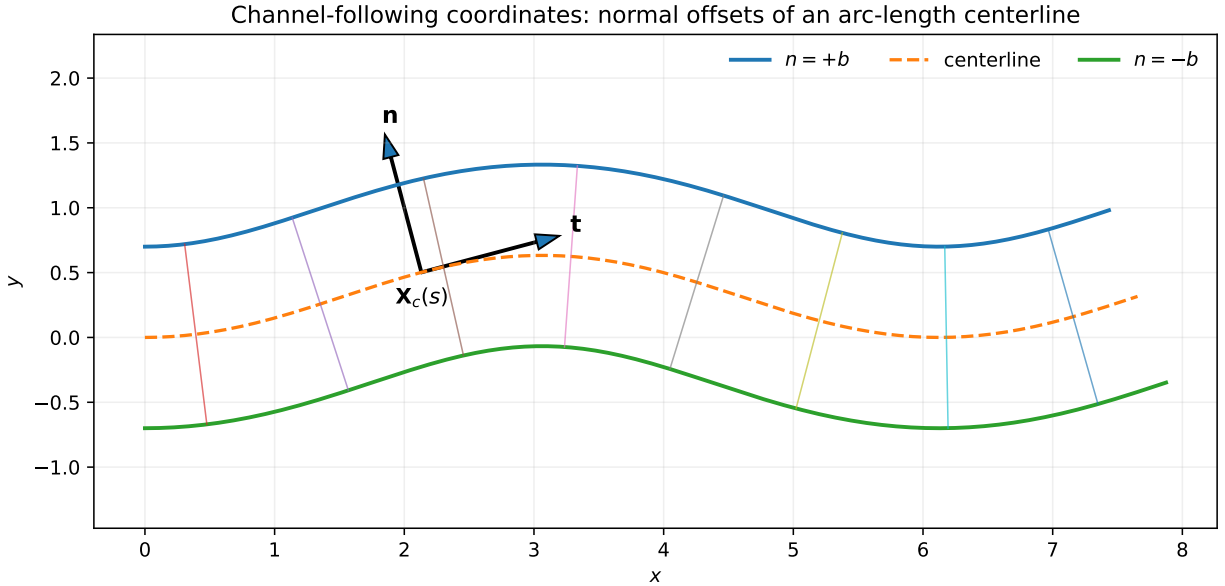


Figure 1: The channel-following map. The coordinate s measures arc length along the centerline; n measures signed physical distance in the local normal direction. The banks $n = \pm b$ are exact normal offsets.

2.3 Geometric validity

The coordinate map is locally nonsingular only when

$$h(s, n) = 1 - \kappa(s)n > 0 \quad (14)$$

throughout the channel. A sufficient condition is

$$b \max_s |\kappa(s)| < 1. \quad (15)$$

This condition prevents neighboring normal coordinate lines from collapsing at the local center of curvature. It does not by itself prevent distant parts of a strongly meandering channel from intersecting; global self-intersection must also be checked when constructing the banks.

Banks generated from $\kappa(s) = \kappa_0 \cos(Ks)$ with $K = 1$

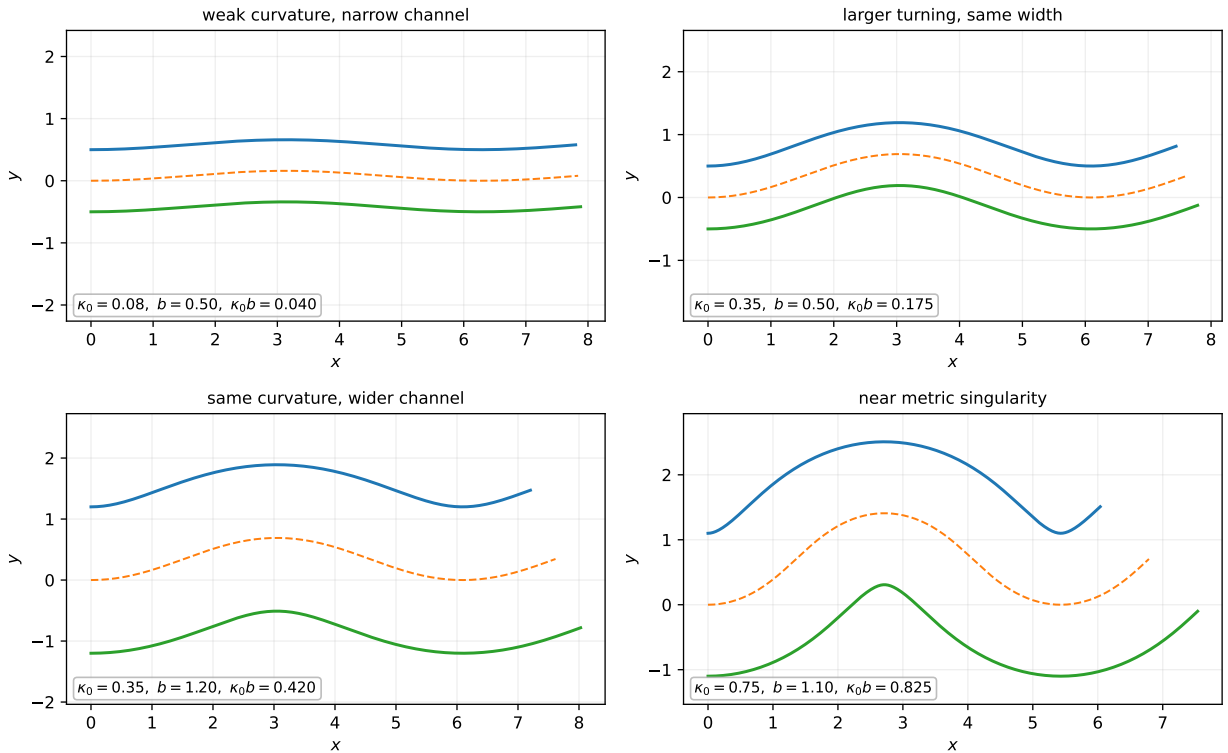


Figure 2: Examples generated by prescribing $\kappa(s) = \kappa_0 \cos(Ks)$ and integrating the tangent angle and centerline. Increasing κ_0 increases turning; increasing b at fixed κ_0 moves the banks closer to the metric-singularity threshold $|\kappa_0|b = 1$. All quantities in this illustrative figure are nondimensional.

3 Exact differential operators

All operators in this section follow from the orthogonal scale factors $h_s = h$ and $h_n = 1$.

3.1 Gradient

For a scalar $a(s, n)$,

$$grada = \frac{1}{h} a_s \mathbf{t} + a_n \mathbf{n}. \quad (16)$$

Here and below, subscripts denote partial derivatives, for example $a_s = \partial a / \partial s$ and $\kappa_s = \partial \kappa / \partial s$.

3.2 Divergence

For a vector written in physical unit-vector components,

$$\mathbf{u} = u_s \mathbf{t} + u_n \mathbf{n}, \quad (17)$$

the divergence is

$$\nabla \cdot \mathbf{u} = \frac{1}{h} [\partial_s u_s + \partial_n (h u_n)]. \quad (18)$$

3.3 Streamfunction representation

Adopt the same sign convention as the motivating Cartesian equation,

$$u_s = -\psi_n, \quad u_n = \frac{1}{h} \psi_s. \quad (19)$$

Substituting Equation (19) into Equation (18) gives

$$\nabla \cdot \mathbf{u} = \frac{1}{h} [-\psi_{ns} + \psi_{sn}] = 0, \quad (20)$$

so incompressibility is satisfied identically.

3.4 Vorticity and the curved Laplacian

The vertical component of relative vorticity is

$$\zeta = (\nabla \times \mathbf{u}) \cdot \mathbf{e}_z = \frac{1}{h} [\partial_s u_n - \partial_n (h u_s)]. \quad (21)$$

Using Equation (19),

$$\zeta = \frac{1}{h} \partial_s \left(\frac{1}{h} \psi_s \right) + \frac{1}{h} \partial_n (h \psi_n). \quad (22)$$

Thus

$$\zeta = \nabla_g^2 \psi \equiv \frac{1}{h} \partial_s \left(\frac{1}{h} \psi_s \right) + \frac{1}{h} \partial_n (h \psi_n). \quad (23)$$

To expand this operator, note from $h = 1 - \kappa n$ that

$$h_s = -n \kappa_s, \quad h_n = -\kappa, \quad (24)$$

$$\left(\frac{1}{h} \right)_s = \frac{n \kappa_s}{h^2}, \quad \left(\frac{1}{h} \right)_n = \frac{\kappa}{h^2}. \quad (25)$$

Therefore

$$\nabla_g^2 \psi = \frac{1}{h^2} \psi_{ss} + \psi_{nn} + \frac{n \kappa_s}{h^3} \psi_s - \frac{\kappa}{h} \psi_n. \quad (26)$$

The two explicitly geometric terms are proportional to κ_s and κ . The factors h^{-2} and h^{-3} also contain curvature to all orders.

3.5 Scalar advection and Jacobian

For any scalar q ,

$$\mathbf{u} \cdot \nabla q = u_s \frac{q_s}{h} + u_n q_n \quad (27)$$

$$= -\psi_n \frac{q_s}{h} + \frac{\psi_s}{h} q_n. \quad (28)$$

Define the curved-coordinate Jacobian

$$J_g(a, b) = \frac{1}{h} (a_s b_n - a_n b_s). \quad (29)$$

Then

$$\mathbf{u} \cdot \nabla q = J_g(\psi, q). \quad (30)$$

The conservative transport form follows from incompressibility:

$$h J_g(\psi, q) = \partial_s(-q\psi_n) + \partial_n(q\psi_s). \quad (31)$$

Equation (31) is useful for conservation diagnostics and finite-volume interpretations.

4 Exact nonlinear barotropic PV equation

For the inviscid barotropic model,

$$q = \zeta = \nabla_g^2 \psi. \quad (32)$$

With linear vorticity drag r and optional forcing F ,

$$q_t + J_g(\psi, q) = -rq + F, \quad q = \nabla_g^2 \psi. \quad (33)$$

In conservative form,

$$\partial_t(hq) + \partial_s(-q\psi_n) + \partial_n(q\psi_s) = h(-rq + F). \quad (34)$$

The geometry is fixed, so $h_t = 0$. If the centerline were allowed to evolve, the time derivative and fluxes would require an arbitrary-Lagrangian–Eulerian treatment; that is outside the present fixed-channel model.

5 Prescribed base flow and its exact vorticity

5.1 Along-channel velocity profile

Assume a base flow tangent to the coordinate lines,

$$\bar{u}_s = U(n), \quad \bar{u}_n = 0. \quad (35)$$

A streamfunction depending only on n produces this flow:

$$\bar{u}_s = -\bar{\psi}_n = U(n), \quad (36)$$

so

$$\bar{\psi}(n) = -\int^n U(\eta) d\eta + C. \quad (37)$$

For the parabolic profile in the motivating figure,

$$\boxed{U(n) = U_0 + \Delta \left(1 - \frac{n^2}{b^2}\right)}, \quad (38)$$

we have

$$\bar{\psi}(n) = -(U_0 + \Delta)n + \frac{\Delta}{3b^2}n^3 + C. \quad (39)$$

5.2 Exact base vorticity

Because $\bar{\psi}_s = 0$, Equation (26) gives

$$\bar{\zeta} = \bar{\psi}_{nn} - \frac{\kappa}{h}\bar{\psi}_n \quad (40)$$

$$= -U_n + \frac{\kappa U}{h}. \quad (41)$$

Thus

$$\boxed{\bar{q} = \bar{\zeta} = -U_n + \frac{\kappa U}{h}}. \quad (42)$$

The first term is the usual Cartesian shear vorticity. The second term is the vorticity of a flow following curved coordinate lines.

5.3 Cross-channel derivative of the base PV

Differentiate Equation (42) with respect to n . Since κ is independent of n ,

$$\bar{q}_n = -U_{nn} + \partial_n \left(\frac{\kappa U}{h} \right) \quad (43)$$

$$= -U_{nn} + \frac{\kappa U_n}{h} + \kappa U \left(\frac{1}{h} \right)_n \quad (44)$$

$$= -U_{nn} + \frac{\kappa U_n}{h} + \frac{\kappa^2 U}{h^2}. \quad (45)$$

Therefore

$$\boxed{\bar{q}_n = -U_{nn} + \frac{\kappa U_n}{h} + \frac{\kappa^2 U}{h^2}}. \quad (46)$$

5.4 Along-channel derivative of the base PV

Although U and $\bar{\psi}$ are independent of s , the base vorticity is not independent of s when curvature varies. Differentiate Equation (42):

$$\bar{q}_s = \partial_s \left(\frac{\kappa U}{h} \right) \quad (47)$$

$$= \frac{\kappa_s U}{h} + \kappa U \left(\frac{1}{h} \right)_s \quad (48)$$

$$= \frac{\kappa_s U}{h} + \frac{n \kappa \kappa_s U}{h^2} \quad (49)$$

$$= \frac{\kappa_s U (h + n \kappa)}{h^2}. \quad (50)$$

Because $h + n\kappa = 1$,

$$\boxed{\bar{q}_s = \frac{\kappa_s U}{h^2}}. \quad (51)$$

This identity is one of the main differences between a constant-curvature bend and a true meander.

5.5 Is the prescribed base flow an exact steady solution?

The nonlinear advection of the base PV is

$$J_g(\bar{\psi}, \bar{q}) = \frac{1}{h} (\bar{\psi}_s \bar{q}_n - \bar{\psi}_n \bar{q}_s) \quad (52)$$

$$= \frac{U}{h} \bar{q}_s \quad (53)$$

$$= \boxed{\frac{\kappa_s U^2}{h^3}}. \quad (54)$$

Hence a profile $U(n)$ in a channel with $\kappa_s \neq 0$ is generally not an unforced steady solution of the inviscid vorticity equation. A numerical model must make one of the following interpretations explicit:

1. impose a mean forcing that maintains $U(n)$;
2. solve for a dynamically consistent two-dimensional base state $\bar{\psi}(s, n)$;
3. assume weak, slowly varying curvature and treat Equation (54) as a small prescribed imbalance.

The perturbation equation below assumes that whatever forcing maintains the base state has been subtracted, so only perturbation dynamics remain.

6 Exact linearized equation with all metric terms retained

6.1 Perturbation decomposition

Write

$$\psi = \bar{\psi} + \psi', \quad q = \bar{q} + q', \quad q' = \nabla_g^2 \psi'. \quad (55)$$

Substitute into the nonlinear equation and subtract the maintained base-state balance. Neglecting the quadratic perturbation term $J_g(\psi', q')$ gives

$$q'_t + J_g(\bar{\psi}, q') + J_g(\psi', \bar{q}) = -r q'. \quad (56)$$

The first Jacobian is

$$J_g(\bar{\psi}, q') = \frac{1}{h} (\bar{\psi}_s q'_n - \bar{\psi}_n q'_s) \quad (57)$$

$$= \frac{U}{h} q'_s. \quad (58)$$

The second is

$$J_g(\psi', \bar{q}) = \frac{1}{h} (\psi'_s \bar{q}_n - \psi'_n \bar{q}_s). \quad (59)$$

Combining Equations (56)–(59),

$$\boxed{q'_t + \frac{U}{h} q'_s + \frac{\bar{q}_n}{h} \psi'_s - \frac{\bar{q}_s}{h} \psi'_n = -r q'}. \quad (60)$$

This compact equation is exact for the fixed geometry and prescribed base profile.

6.2 Expanded exact equation for general $U(n)$

Substitute Equations (46) and (51) into Equation (60):

$$\boxed{q'_t + \frac{U}{h}q'_s + \left[-\frac{U_{nn}}{h} + \frac{\kappa U_n}{h^2} + \frac{\kappa^2 U}{h^3}\right]\psi'_s - \frac{\kappa_s U}{h^3}\psi'_n = -rq'}. \quad (61)$$

The perturbation PV inversion is

$$\boxed{q' = \frac{1}{h^2}\psi'_{ss} + \psi'_{nn} + \frac{n\kappa_s}{h^3}\psi'_s - \frac{\kappa}{h}\psi'_n}. \quad (62)$$

Equations (61) and (62), together with $h = 1 - \kappa n$, are the exact all-terms-retained linear model.

6.3 Specialization to the parabolic profile

For Equation (38),

$$U_n = -\frac{2\Delta n}{b^2}, \quad U_{nn} = -\frac{2\Delta}{b^2}. \quad (63)$$

Equation (61) becomes

$$\boxed{q'_t + \frac{U}{h}q'_s + \left[\frac{2\Delta}{b^2 h} - \frac{2\Delta \kappa n}{b^2 h^2} + \frac{\kappa^2 U}{h^3}\right]\psi'_s - \frac{\kappa_s U}{h^3}\psi'_n = -rq'}. \quad (64)$$

In the straight-channel limit $\kappa = \kappa_s = 0$ and $h = 1$,

$$q'_t + Uq'_s + \frac{2\Delta}{b^2}\psi'_s = -rq', \quad q' = \psi'_{ss} + \psi'_{nn}, \quad (65)$$

which recovers the equation shown in the motivating figure.

7 Sinusoidal curvature: exact geometry and exact equation

7.1 Prescribing curvature directly

Let

$$\boxed{\kappa(s) = \kappa_0 \cos(Ks)}. \quad (66)$$

Then

$$\boxed{\kappa_s = -K\kappa_0 \sin(Ks)}, \quad (67)$$

and

$$\boxed{h(s, n) = 1 - n\kappa_0 \cos(Ks)}. \quad (68)$$

The tangent angle is found from $\theta_s = \kappa$:

$$\theta(s) = \theta_0 + \frac{\kappa_0}{K} \sin(Ks). \quad (69)$$

Taking $\theta_0 = 0$, the exact centerline is obtained by quadrature,

$$x_c(s) = x_c(0) + \int_0^s \cos \left[\frac{\kappa_0}{K} \sin(K\sigma) \right] d\sigma, \quad (70)$$

$$y_c(s) = y_c(0) + \int_0^s \sin \left[\frac{\kappa_0}{K} \sin(K\sigma) \right] d\sigma. \quad (71)$$

The exact banks are

$$x_{\pm}(s) = x_c(s) \mp b \sin \theta(s), \quad (72)$$

$$y_{\pm}(s) = y_c(s) \pm b \cos \theta(s). \quad (73)$$

7.2 Exact perturbation PV inversion

Substitute Equations (66)–(68) into Equation (62):

$$\begin{aligned} q' &= \frac{1}{h^2} \psi'_{ss} + \psi'_{nn} - \frac{nK\kappa_0 \sin(Ks)}{h^3} \psi'_s - \frac{\kappa_0 \cos(Ks)}{h} \psi'_n, \\ h &= 1 - n\kappa_0 \cos(Ks). \end{aligned} \quad (74)$$

7.3 Exact linear equation for the parabolic profile

Equation (64) becomes

$$\begin{aligned} & q'_t + \frac{U}{h} q'_s \\ & + \left[\frac{2\Delta}{b^2 h} - \frac{2\Delta\kappa_0 n \cos(Ks)}{b^2 h^2} + \frac{\kappa_0^2 \cos^2(Ks) U}{h^3} \right] \psi'_s \\ & + \frac{UK\kappa_0 \sin(Ks)}{h^3} \psi'_n = -rq'. \end{aligned} \quad (75)$$

Equations (74) and (75) are the exact periodic-meander linear model with all metric terms retained.

8 Weak-curvature approximation

8.1 Ordering parameter

Define

$$\epsilon_c = b \max_s |\kappa(s)|. \quad (76)$$

For sinusoidal curvature, $\epsilon_c = b|\kappa_0|$. The weak-curvature approximation assumes

$$\epsilon_c \ll 1. \quad (77)$$

Since $|n| \leq b$, this ensures $|\kappa n| \ll 1$ throughout the domain. We retain terms linear in κ and κ_s and discard products such as κ^2 , $\kappa\kappa_s$, and higher powers.

The metric factors expand as

$$\frac{1}{h} = 1 + \kappa n + \mathcal{O}(\epsilon_c^2), \quad (78)$$

$$\frac{1}{h^2} = 1 + 2\kappa n + \mathcal{O}(\epsilon_c^2), \quad (79)$$

$$\frac{1}{h^3} = 1 + 3\kappa n + \mathcal{O}(\epsilon_c^2). \quad (80)$$

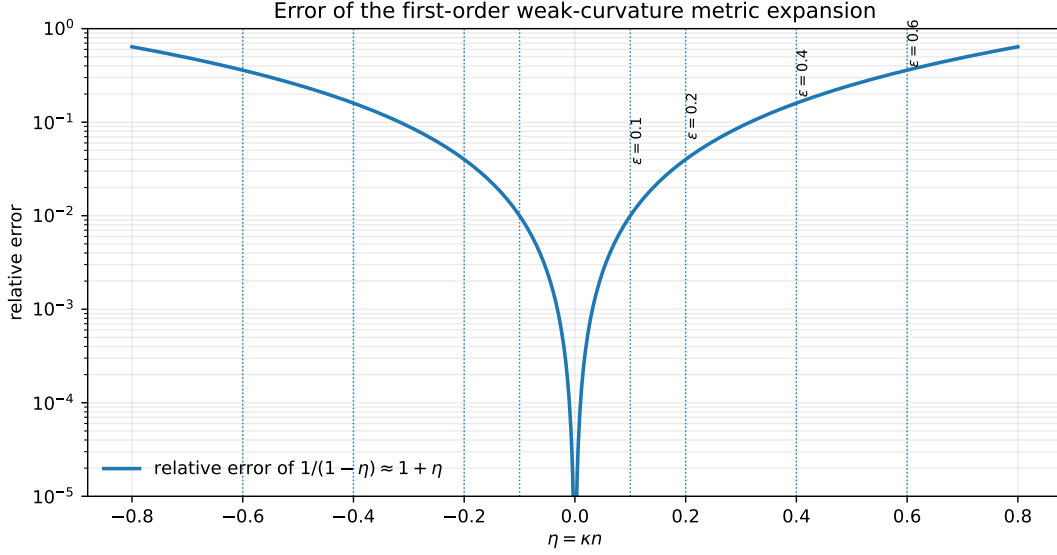


Figure 3: Relative error in the first-order approximation $(1 - \eta)^{-1} \approx 1 + \eta$, where $\eta = \kappa n$. The error is exactly η^2 for this particular factor. Similar first-order truncation errors scale as $\mathcal{O}[(\kappa n)^2]$.

8.2 Weak-curvature PV inversion

Start from Equation (62):

$$q' = h^{-2}\psi'_{ss} + \psi'_{nn} + n\kappa_s h^{-3}\psi'_s - \kappa h^{-1}\psi'_n. \quad (81)$$

Apply Equation (80). In the κ_s term, the correction $3\kappa n$ would produce a product $\kappa\kappa_s$ and is discarded. In the last term, multiplying $-\kappa$ by the κn correction would produce κ^2 and is discarded. Therefore

$$q' = \psi'_{ss} + \psi'_{nn} + 2\kappa n\psi'_{ss} + n\kappa_s\psi'_s - \kappa\psi'_n + \mathcal{O}(\epsilon_c^2). \quad (82)$$

It is useful to separate the straight and curvature operators:

$$q' = q'_0 + q'_1 + \mathcal{O}(\epsilon_c^2), \quad (83)$$

with

$$q'_0 = \psi'_{ss} + \psi'_{nn}, \quad (84)$$

$$q'_1 = 2\kappa n\psi'_{ss} + n\kappa_s\psi'_s - \kappa\psi'_n. \quad (85)$$

8.3 Weak-curvature evolution equation for general $U(n)$

Begin with the exact Equation (61). Expand each coefficient:

$$\frac{U}{h} = U(1 + \kappa n) + \mathcal{O}(\epsilon_c^2), \quad (86)$$

$$-\frac{U_{nn}}{h} + \frac{\kappa U_n}{h^2} + \frac{\kappa^2 U}{h^3} = -U_{nn}(1 + \kappa n) + \kappa U_n + \mathcal{O}(\epsilon_c^2) \quad (87)$$

$$= -U_{nn} + \kappa(U_n - nU_{nn}) + \mathcal{O}(\epsilon_c^2), \quad (88)$$

and

$$-\frac{\kappa_s U}{h^3} = -\kappa_s U + \mathcal{O}(\epsilon_c^2). \quad (89)$$

Hence

$$\begin{aligned} & q'_t + U(1 + \kappa n)q'_s \\ & + [-U_{nn} + \kappa(U_n - nU_{nn})]\psi'_s - U\kappa_s\psi'_n = -rq' + \mathcal{O}(\epsilon_c^2). \end{aligned} \quad (90)$$

8.4 Cancellation for the parabolic profile

For the parabolic profile,

$$U_n = -\frac{2\Delta n}{b^2}, \quad U_{nn} = -\frac{2\Delta}{b^2}. \quad (91)$$

Therefore

$$U_n - nU_{nn} = -\frac{2\Delta n}{b^2} + n\frac{2\Delta}{b^2} = 0. \quad (92)$$

The first-order correction to the base cross-channel PV-gradient coefficient cancels exactly. Equation (90) simplifies to

$$q'_t + U(1 + \kappa n)q'_s + \frac{2\Delta}{b^2}\psi'_s - U\kappa_s\psi'_n = -rq' + \mathcal{O}(\epsilon_c^2). \quad (93)$$

Together with Equation (82), this is the principal weak-curvature model.

For the chosen parabolic $U(n)$, the exact equation contains four distinct geometric modifications: h^{-1} advection, two $\mathcal{O}(\kappa)$ corrections inside the base-PV-gradient coefficient, a κ^2 term, and a κ_s term. At first order, the two $\mathcal{O}(\kappa)$ base-PV-gradient corrections cancel, the κ^2 term is discarded, and only the modified advection and κ_s coupling remain explicitly in the evolution equation. Curvature also remains in the PV inversion operator.

8.5 Straight-operator plus explicit curvature forcing

Equation (93) can be rearranged as

$$\begin{aligned} q'_t + Uq'_s + \frac{2\Delta}{b^2}\psi'_s = & -rq' \\ & - U\kappa n q'_s + U\kappa_s\psi'_n + \mathcal{O}(\epsilon_c^2). \end{aligned} \quad (94)$$

This form is attractive numerically because the straight-channel operator can remain on the implicit left-hand side, while the curvature terms can be evaluated explicitly.

9 Weak-curvature form for sinusoidal curvature

For

$$\kappa = \kappa_0 \cos(Ks), \quad \kappa_s = -K\kappa_0 \sin(Ks), \quad (95)$$

Equation (82) becomes

$$\begin{aligned} q' = & \psi'_{ss} + \psi'_{nn} + 2\kappa_0 n \cos(Ks)\psi'_{ss} \\ & - K\kappa_0 n \sin(Ks)\psi'_s - \kappa_0 \cos(Ks)\psi'_n + \mathcal{O}(\epsilon_c^2). \end{aligned} \quad (96)$$

The parabolic-profile evolution equation is

$$\begin{aligned} q'_t + U[1 + \kappa_0 n \cos(Ks)]q'_s + \frac{2\Delta}{b^2}\psi'_s \\ + UK\kappa_0 \sin(Ks)\psi'_n = -rq' + \mathcal{O}(\epsilon_c^2). \end{aligned} \quad (97)$$

Equivalently,

$$\boxed{\begin{aligned} q'_t + Uq'_s + \frac{2\Delta}{b^2}\psi'_s &= -rq' \\ &\quad - U\kappa_0 n \cos(Ks)q'_s \\ &\quad - UK\kappa_0 \sin(Ks)\psi'_n + \mathcal{O}(\epsilon_c^2). \end{aligned}} \quad (98)$$

The sign in the final term follows from $\kappa_s = -K\kappa_0 \sin(Ks)$ and the rearrangement of $-U\kappa_s\psi'_n$ from the left-hand side to the right-hand side.

10 Small slope, weak curvature, and slow variation

These are separate assumptions.

10.1 Weak curvature

The metric expansion is controlled by

$$\boxed{\epsilon_c = b|\kappa_0| \ll 1.} \quad (99)$$

It compares the channel half-width with the minimum radius of curvature. This assumption is required for Equations (82)–(98).

10.2 Small centerline slope

For the prescribed curvature,

$$\theta(s) = \frac{\kappa_0}{K} \sin(Ks), \quad (100)$$

so the maximum tangent-angle magnitude is

$$\boxed{\epsilon_s = \frac{|\kappa_0|}{K}.} \quad (101)$$

Small slope means

$$\boxed{\epsilon_s \ll 1,} \quad (102)$$

because

$$\frac{dy_c}{dx_c} = \tan \theta \approx \theta. \quad (103)$$

Under this additional approximation, $s \approx x$ and

$$\boxed{y_c(x) \approx -\frac{\kappa_0}{K^2} \cos(Kx) + \text{constant}.} \quad (104)$$

Small slope is not required when the exact arc-length coordinate construction is used.

10.3 Slowly varying curvature

The variation length scale is $L_\kappa \sim K^{-1}$. Compare it with the channel half-width using

$$\boxed{\epsilon_v = Kb.} \quad (105)$$

If $Kb \ll 1$, curvature varies little over a cross-channel distance. The combination

$$b^2 \max |\kappa_s| = Kb(b|\kappa_0|) = \epsilon_v \epsilon_c \quad (106)$$

controls the size of some κ_s effects after nondimensionalization. The first-order weak-curvature equations retain κ_s and therefore do not require $Kb \ll 1$. Slow variation is needed only if one also wants to neglect the κ_s terms or make a locally constant-curvature approximation.

It is possible to have weak curvature but not small slope. For example, a very long meander can have a large cumulative turning angle while its radius of curvature remains much larger than the channel width. Conversely, a nearly horizontal centerline can still have curvature that is not weak relative to a very wide channel.

11 Boundary conditions

11.1 Impermeable banks

At a fixed bank $n = \pm b$, no normal flow requires

$$u_n = \frac{1}{h} \psi_s = 0. \quad (107)$$

Therefore ψ is constant along each bank:

$$\psi(s, +b, t) = C_+(t), \quad \psi(s, -b, t) = C_-(t). \quad (108)$$

The transport across a cross-section is

$$\mathcal{Q} = \int_{-b}^b u_s \, dn = - \int_{-b}^b \psi_n \, dn \quad (109)$$

$$= \psi(s, -b, t) - \psi(s, +b, t) = C_-(t) - C_+(t). \quad (110)$$

For perturbations at fixed total discharge, a convenient choice is

$$\boxed{\psi'(s, -b, t) = 0, \quad \psi'(s, +b, t) = 0.} \quad (111)$$

Only the difference between the two bank constants has physical significance; an overall additive constant is a streamfunction gauge.

11.2 Streamwise periodicity

For a repeating meander with period $L_s = 2\pi/K$,

$$\boxed{\psi'(s + L_s, n, t) = \psi'(s, n, t), \quad q'(s + L_s, n, t) = q'(s, n, t).} \quad (112)$$

The curvature and metric coefficients are periodic with the same period.

11.3 If lateral viscosity is added

For the inviscid equation or Rayleigh drag alone, Equation (111) is sufficient for the second-order PV inversion. If a physical lateral viscosity is introduced, an additional wall-stress condition is needed. With the present sign convention:

No slip.

$$\psi' = 0, \quad \psi'_n = 0 \quad \text{at } n = \pm b. \quad (113)$$

Zero tangential stress. For an orthogonal curved wall, the shear-strain condition is

$$\partial_n u_s + \frac{\kappa}{h} u_s = 0. \quad (114)$$

Using $u_s = -\psi_n$,

$$\boxed{\psi_{nn} + \frac{\kappa}{h} \psi_n = 0 \quad \text{at } n = \pm b.} \quad (115)$$

This is not generally equivalent to setting $\zeta = 0$ at a curved wall.

12 Numerical formulation in a rectangular computational domain

The coordinate transformation maps the physical meandering channel to

$$0 \leq s < L_s, \quad -b \leq n \leq b. \quad (116)$$

A natural spectral discretization is Fourier in s and Chebyshev in n .

12.1 Exact all-terms-retained model

Use prognostic q' and diagnostic ψ' :

$$q'_t + \frac{U}{h} q'_s + \left[-\frac{U_{nn}}{h} + \frac{\kappa U_n}{h^2} + \frac{\kappa^2 U}{h^3} \right] \psi'_s - \frac{\kappa_s U}{h^3} \psi'_n = -r q', \quad (117)$$

$$q' = \frac{1}{h^2} \psi'_{ss} + \psi'_{nn} + \frac{n \kappa_s}{h^3} \psi'_s - \frac{\kappa}{h} \psi'_n. \quad (118)$$

For sinusoidal curvature, all coefficients are known periodic functions of (s, n) . The cost is that two-dimensional nonconstant coefficients couple streamwise Fourier modes, especially if placed on the implicit side of the solve.

12.2 Weak-curvature split

A computationally efficient first model keeps the straight inversion as the principal operator:

$$q'_0 = \psi'_{ss} + \psi'_{nn}, \quad (119)$$

and evaluates

$$q'_1 = 2\kappa n \psi'_{ss} + n \kappa_s \psi'_s - \kappa \psi'_n \quad (120)$$

as an explicit correction. The evolution equation can be written in the split form of Equation (94). This preserves a constant-coefficient elliptic core while retaining every first-order curvature effect.

There are two possible implementations of the perturbation PV:

1. define $q' = q'_0 + q'_1$ diagnostically at each stage;
2. evolve the exact first-order q' and solve the weak variable-coefficient inversion.

The first is simpler and cheaper but should be verified against the second and against the exact metric model at selected parameters.

12.3 Recommended verification sequence

1. Set $\kappa_0 = 0$ and reproduce the straight-channel dispersion relation or time evolution.
2. Set $\kappa = \kappa_0$ constant, so $\kappa_s = 0$, and compare the numerical Laplacian against analytic annular-coordinate identities.
3. Use a small sinusoidal $\kappa_0 b$ and verify that the difference between exact and weak solutions scales approximately as $(\kappa_0 b)^2$.
4. Increase $\kappa_0 b$ until the weak model visibly departs from the exact model.
5. Monitor the minimum metric factor

$$h_{\min} = 1 - b|\kappa_0|. \quad (121)$$

Do not approach $h_{\min} = 0$ in a spectral calculation.

13 Extensions to a more general QGPV

Suppose

$$q = \nabla_g^2 \psi - \frac{\psi}{L_D^2} + Q(s, n), \quad (122)$$

where Q may represent planetary or topographic background PV. The nonlinear evolution remains

$$q_t + J_g(\psi, q) = -r \nabla_g^2 \psi + F, \quad (123)$$

with the drag form chosen according to the physical model.

For the same base streamfunction $\bar{\psi}(n)$,

$$\bar{q} = -U_n + \frac{\kappa U}{h} - \frac{\bar{\psi}}{L_D^2} + Q. \quad (124)$$

The derivatives entering the linear equation become

$$\boxed{\bar{q}_n = -U_{nn} + \frac{\kappa U_n}{h} + \frac{\kappa^2 U}{h^2} + \frac{U}{L_D^2} + Q_n,} \quad (125)$$

$$\boxed{\bar{q}_s = \frac{\kappa_s U}{h^2} + Q_s.} \quad (126)$$

The compact exact linear equation retains the same structure,

$$q'_t + \frac{U}{h} q'_s + \frac{\bar{q}_n}{h} \psi'_s - \frac{\bar{q}_s}{h} \psi'_n = \text{dissipation}, \quad (127)$$

while the inversion is

$$q' = \nabla_g^2 \psi' - \frac{\psi'}{L_D^2}. \quad (128)$$

14 Consistency checks

14.1 Straight-channel limit

Setting $\kappa = 0$ gives $h = 1$, $\kappa_s = 0$, and

$$\nabla_g^2 \psi = \psi_{ss} + \psi_{nn}. \quad (129)$$

The exact linear equation reduces to

$$q'_t + Uq'_s - U_{nn}\psi'_s = -rq'. \quad (130)$$

For the parabolic profile, $-U_{nn} = 2\Delta/b^2$.

14.2 Constant-curvature limit

If $\kappa = \kappa_0$ is constant, then $\kappa_s = 0$. All terms involving along-channel changes of curvature vanish, but the exact metric factors and the curvature contribution to the base PV gradient remain. This is the annular-channel test case.

14.3 Dimensional check

The relevant dimensions are

$$[s] = [n] = L, \quad [\kappa] = L^{-1}, \quad [\kappa_s] = L^{-2}, \quad [\psi] = L^2 T^{-1}, \quad [q] = T^{-1}. \quad (131)$$

Each term in the evolution equation has dimension T^{-2} .

14.4 Metric-expansion error

For the basic factor,

$$\frac{1}{1-\eta} - (1+\eta) = \frac{\eta^2}{1-\eta}, \quad (132)$$

so its relative error is exactly η^2 . Since $|\eta| = |\kappa n| \leq \epsilon_c$, the first-order metric error is at most of order ϵ_c^2 away from the singular limit. This is an operator-level estimate; solution errors can be amplified near marginal instabilities or resonances.

15 Final equations for direct use

15.1 Exact sinusoidal-curvature model

Let

$$\kappa = \kappa_0 \cos(Ks), \quad \kappa_s = -K\kappa_0 \sin(Ks), \quad h = 1 - \kappa n, \quad (133)$$

$$U = U_0 + \Delta \left(1 - \frac{n^2}{b^2} \right). \quad (134)$$

Then

$$q' = \frac{1}{h^2} \psi'_{ss} + \psi'_{nn} + \frac{n\kappa_s}{h^3} \psi'_s - \frac{\kappa}{h} \psi'_n, \quad (135)$$

$$\begin{aligned} q'_t + \frac{U}{h} q'_s + \left[\frac{2\Delta}{b^2 h} - \frac{2\Delta \kappa n}{b^2 h^2} + \frac{\kappa^2 U}{h^3} \right] \psi'_s \\ - \frac{\kappa_s U}{h^3} \psi'_n = -rq'. \end{aligned} \quad (136)$$

15.2 First-order weak-curvature model

For $b|\kappa_0| \ll 1$,

$$q' = \psi'_{ss} + \psi'_{nn} + 2\kappa n \psi'_{ss} + n \kappa_s \psi'_s - \kappa \psi'_n, \quad (137)$$

$$q'_t + U(1 + \kappa n)q'_s + \frac{2\Delta}{b^2} \psi'_s - U \kappa_s \psi'_n = -r q'. \quad (138)$$

The boundary conditions for a fixed-discharge perturbation problem are

$$\psi'(s, \pm b, t) = 0, \quad \psi'(s + 2\pi/K, n, t) = \psi'(s, n, t). \quad (139)$$

The exact and weak equations use the same rectangular computational domain and the same wall conditions. The weak model mainly simplifies the nonconstant metric factors and removes all second-order curvature contributions. It does not remove curvature from the PV inversion, and it does not justify replacing arc length by Cartesian x unless the independent small-slope condition $|\kappa_0|/K \ll 1$ also holds.

A Curvature of a Cartesian cosine centerline

If instead the centerline is prescribed as

$$y_c(x) = A \cos(kx), \quad (140)$$

then

$$y'_c = -Ak \sin(kx), \quad y''_c = -Ak^2 \cos(kx). \quad (141)$$

The exact signed curvature is

$$\kappa(x) = \frac{-Ak^2 \cos(kx)}{[1 + A^2 k^2 \sin^2(kx)]^{3/2}}. \quad (142)$$

Because

$$\frac{ds}{dx} = \sqrt{1 + y'^2_c}, \quad (143)$$

we have

$$\kappa_s = \frac{1}{\sqrt{1 + y'^2_c}} \frac{d\kappa}{dx}. \quad (144)$$

When $Ak \ll 1$,

$$s \approx x, \quad \kappa \approx -Ak^2 \cos(kx), \quad \kappa_s \approx Ak^3 \sin(kx). \quad (145)$$

Comparing this with prescribed sinusoidal curvature gives

$$\kappa_0 \approx -Ak^2, \quad K \approx k. \quad (146)$$

B Sign-convention conversion

This document uses

$$u_s = -\psi_n, \quad u_n = h^{-1} \psi_s, \quad \zeta = \nabla_g^2 \psi. \quad (147)$$

If the opposite streamfunction convention is used, both velocity components and the sign relation between vorticity and the Laplacian change consistently. The safest implementation practice is to verify the straight-channel limit against the original Cartesian equation before enabling curvature.